CPSC 481
Artificial Intelligence

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Group

Syllabus

Week 7: Probability & Uncertainty

Transitioning from Logic to Probability
Textbook: AIMA 4th Ed. Chapter 12
Midterm Reminder: This Thursday!

Recap:

- Heuristic search is the backbone of classical AI.
- Single-agent, static environments.
- Multi-agent environments (Games).
- Compare A* and Greedy on the 8-puzzle.

Acting Under Uncertainty

- Real-world agents deal with partial observability and nondeterminism
- Logical agents struggle with the "Qualification Problem"
- Solution: Numeric degrees of belief

Why Logic Fails for Diagnosis

- Laziness: Too many rules to list
- Theoretical Ignorance: No complete theory of the domain
- Practical Ignorance: Missing data for a specific patient

Probability: The Agent's Tool

- Summarizes uncertainty from laziness and ignorance
- Numerical belief: 0 (certainly false) to 1 (certainly true
- Indistinguishable situations are grouped statistically

Rational Decisions

- Decision Theory = Probability + Utility
- Principle of Maximum Expected Utility (MEU)
- A rational agent chooses the action with the highest average payoff

∑ω P(ω) = 1

Basic Probability Notation

- Sample Space (Ω): Set of all possible worlds
- Probability Model: Associates $P(\omega)$ with each world
- Normalization: $\Sigma\omega$P(ω) = 1

Events and Propositions

- An event is a set of possible worlds
- Probability of a proposition = $\sum P(w)$ where it holds
- Example: $P(Total = 11)$ for two dice

Unconditional (Prior) Probability

- Degrees of belief in the absence of new info
- Represents general knowledge before evidence
- Example: $P(cavity) = 0.2$

Conditional (Posterior) Probability

- Belief after evidence is revealed: $P(a|b)$
- Definition: $P(a|b) = P(a \wedge b) / P(b)$
- Updates belief states as percepts arrive

The Product Rule

- Fundamental identity: $P(a \wedge b) = P(a|b)P(b)$
- For both to be true, b must be true and a must be true given b
- Used extensively in Bayesian derivations

Random Variables

- Variables take values from a predefined range
- Boolean (True/False) or Multi-valued (Sun, Rain, etc.)
- Notation: $P(Weather)$ refers to the entire distribution vector

Probability Distributions

- Discrete: Categorical distribution (e.g., Weather)
- Continuous: Probability Density Functions (PDF)
- Joint Distribution: Probabilities for all combinations of variables

The Full Joint Distribution

- Complete model for all variables in a domain
- Allows calculation of any probabilistic query
- Major drawback: Size grows exponentially $O(2^n)$

Kolmogorov's Axioms

- 1. $0 < P(\omega) > 1$
- 2. $\sum P(\omega) = 1$
- 3. Inclusion-Exclusion: $P(a \vee b) = P(a) + P(b) - P(a \wedge b)$

The Dutch Book Argument

- De Finetti's Theorem: Inconsistent beliefs lead to guaranteed loss
- No rational agent can violate probability axioms
- Every action is essentially a bet on an outcome

Inference by Enumeration

- Querying the Full Joint Distribution
- Marginalization: "Summing out" unobserved variables
- Conditioning: Re-expressing marginals using conditional probabilities

Normalization (The Alpha Shortcut)

- Computing $P(X|e)$ without knowing $P(e)$
- Calculate relative proportions and divide by their sum
- Reduces data assessment requirements

Independence

- Two variables X and Y are independent if $P(X, Y) = P(X)P(Y)$
- Drastically reduces representation size
- Allows factoring the joint distribution into smaller pieces

Bayes' Rule

- Identity: $P(b|a) = [P(a|b)P(b)] / P(a)$
- Underlies most modern AI probabilistic inference
- Standard for causal vs. diagnostic reasoning

Causal vs. Diagnostic Reasoning

- Doctors know $P(Symptom \mid Disease)$ (Causal)
- Doctors need $P(Disease \mid Symptom)$ (Diagnostic)
- Causal knowledge is often more robust to epidemics

Conditional Independence

- X and Y are independent given Z
- Allows scaling up to massive numbers of symptoms
- Key development in modern AI history

Naive Bayes Model

- Assumes all effects are conditionally independent given the cause
- Grows linearly $O(n)$ with number of effects
- Surprisingly effective despite "naive" assumptions

Example: Text Classification

- Cause: Document Category; Effect: Presence of words
- Used for spam filtering and language determination
- Overconfident but accurate in ranking

Homework: Week 7

- 1. Exercises 12.1 and 12.3 from the textbook
- 2. Manual calculation: Apply Bayes' Rule to a medical case
- Due: Next Tuesday at start of class

MIDTERM: This Thursday

- Coverage: Week 1 through Week 6
- Topics: AI History, Agents, PEAS, Search (Uninformed/Heuristic), A*
- Format: Same as Sample Questions 4 in 1 hour.

Conclusion & Looking Ahead

- Heuristic search is the backbone of classical AI.
- Today: Single-agent, static environments.
- Next: Multi-agent environments (Games).
- Assignment: Compare A* and Greedy on the 8-puzzle.
- Lab: Implementation of A* in Python.